

Impact Statement

Domain-Specific Small Language Model for African Folktales

This project explores the development of a domain-specific Small Language Model (SLM) trained on a curated corpus of African folktales to generate short narratives from creative or thematic prompts. Its potential societal value lies in supporting the digital accessibility and continued appreciation of storytelling traditions that may become less accessible as oral transmission declines. By making selected folktale narratives more accessible in digital environments, the project may benefit students, educators, researchers, younger generations, cultural preservation initiatives, and African communities interested in engaging with traditional stories, moral lessons, and cultural knowledge. However, the model should be viewed as a computational exploration of these narratives rather than a replacement for storytellers, communities, or authoritative cultural records.

Educationally, the project could provide culturally relevant material for exploring African literature, storytelling structures, and responsible AI. Technologically, it demonstrates how smaller, domain-specific language models may be applied to underrepresented cultural datasets while potentially requiring fewer computational resources than larger general-purpose models. This may encourage further research into culturally inclusive and resource-efficient AI systems.

Fairness and representation are central concerns. Africa contains substantial cultural, linguistic, regional, and ethnic diversity, and its storytelling traditions cannot responsibly be represented as a single homogeneous tradition. A limited dataset may disproportionately represent certain regions, languages, themes, or communities while excluding others. Such imbalance could cause the model to reproduce stereotypes, blur distinctions between traditions, or present particular narrative patterns as representative of all African traditions. These issues affect dignity, cultural identity, autonomy, and community well-being, particularly where communities have limited influence over how their stories are collected or reused.

The model may also hallucinate traditions, historical information, or cultural details that are inaccurate or culturally insensitive. Converting oral narratives into machine-readable text can remove performance, language, setting, and social context that contribute to their meaning. Copyright, ownership, consent, attribution, and benefit-sharing questions may arise where stories are sourced from living communities or protected collections. Generated stories should therefore be clearly labeled as AI-generated and must not automatically be treated as authentic cultural records.

Responsible development should include expanding and diversifying the corpus; documenting provenance, permissions, and known gaps where possible; and consulting cultural experts, storytellers, educators, and relevant communities. Human evaluation can help identify harmful stereotypes, inaccurate cultural claims, and insensitive outputs. Clear distinctions between documented narratives and generated content, together with ongoing monitoring and revision, are important safeguards.

Over time, this work may contribute to research on culturally inclusive AI and digital cultural preservation. Its present contribution is necessarily limited: neither the dataset nor the model can represent the full diversity, living context, or authority of African storytelling traditions. Any long-term value depends on sustained community participation, transparent stewardship, and careful use alongside, rather than in place of, human cultural knowledge.